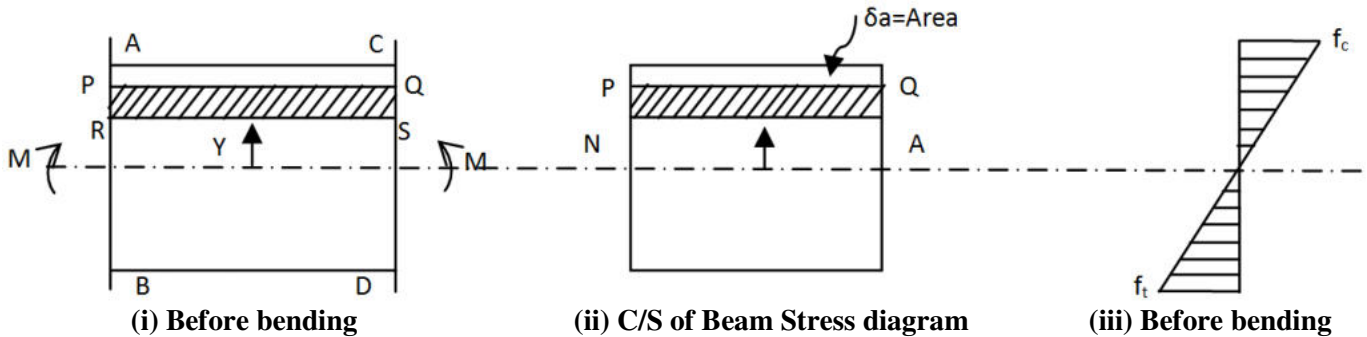


➤ **Equation of bending stress (Flexure eqn)**



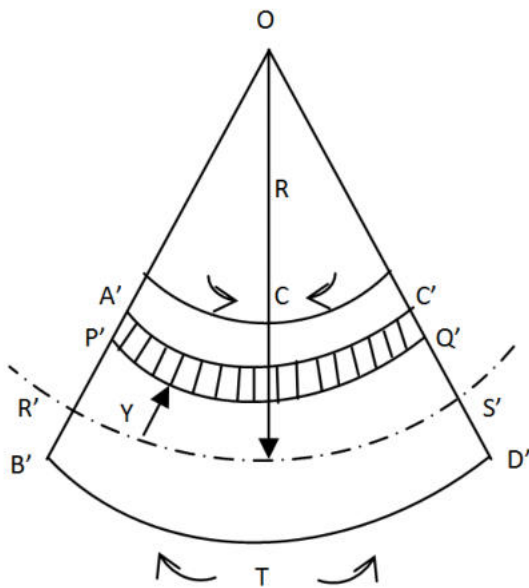
- Consider a layer PQ at a distance Y from the neutral axis.
- after bending layer of PQ compressed to P'Q'

∴ Decrease in length of PQ layer

$$\delta l = PQ - P'Q'$$

$$\therefore \text{Strain } (\varepsilon) = \frac{\delta l}{\text{original length}} = \frac{PQ - P'Q'}{PQ} \quad (1)$$

- Now from geometry of the fig two section OP'Q' & OR'S' are similar.



$$\therefore \frac{P'Q'}{R'S'} = \frac{R-y}{R} \Rightarrow 1 - \frac{P'Q'}{R'S'} = 1 - \frac{R-y}{R}$$

$$\therefore \frac{R'S' - P'Q'}{R'S'} = \frac{R - R + Y}{R} = \frac{Y}{R} \quad \therefore R'S' = RS = PQ = \text{Neutral layer}$$

$$\therefore \frac{PQ - P'Q'}{PQ} = \frac{Y}{R} \quad (2)$$

from eqⁿ (1) & (2)

$$\therefore \varepsilon = \frac{Y}{R} \quad \therefore \text{bending stress } (\sigma) = \varepsilon \times E$$

$$\therefore \frac{\sigma}{E} = \frac{Y}{R} \quad \therefore \varepsilon = \frac{\sigma}{E}$$

$$\therefore \frac{\sigma}{Y} = \frac{E}{R} \quad (3)$$

Now for fig. i.e. C/S of beam, in which PQ small layer is at a distance Y from NA.

δa = Area of PQ layer

As we know that from eqⁿ (3) $\therefore \sigma = Y \times \frac{E}{R}$

$\therefore$ Decrease in length of PQ layer = $\sigma \times \delta a = Y \times \frac{E}{R} \times \delta a$

$\therefore$ Moment of this force about NA

$$= \left(Y \times \frac{E}{R} \times \delta a \right) \times Y = Y^2 \times \frac{E}{R} \times \delta a$$

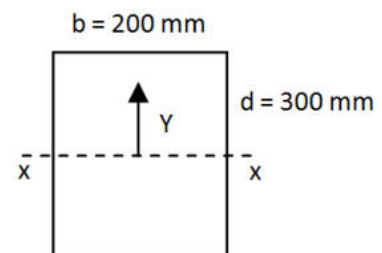
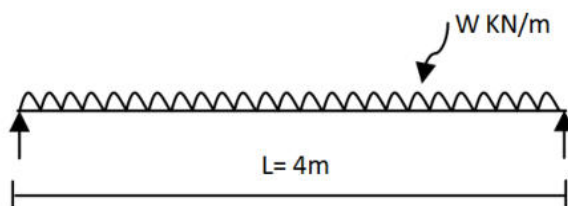
Algebraic sum of all such moment about NA

$$\begin{aligned} M &= \sum \frac{E}{R} \times Y^2 \times \delta a \\ &= \frac{E}{R} \sum Y^2 \times \delta a \\ &= \frac{E}{R} \times I \quad \left(\because I = \sum Y^2 \times \delta a \right) \end{aligned}$$

Now, from eqⁿ (3)

$$\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$$

Example 1 A simply supported beam of span 4.0 has a cross-section 200mm x 300mm. If the permissible stress in the material of the beam is 20 N/mm^2 , determine maximum udl it can carry.



Answer:

$$I_{xx} = \frac{bd^3}{12} = \frac{200 \times 300^3}{12} = 450 \times 10^6 \text{ mm}^4$$

$$\sigma = 20 \text{ N/mm}^2$$

$$y = \frac{d}{2} = \frac{300}{2} = 150 \text{ mm}$$

$$\text{flexure eq}^n \quad \frac{M}{I} = \frac{\sigma}{Y}$$

$$\therefore M = \frac{\sigma \times I}{Y} = \frac{20 \times 450 \times 10^6}{150} = 600 \times 10^6 \text{ N} \cdot \text{mm}$$

Now, for simply supported beam with U.D.L on entire span

$$\therefore M_{\max} = \frac{wl^2}{8} = 600 \times 10^6$$

$$\therefore \frac{w \times (4000)^2}{8} = 600 \times 10^6$$

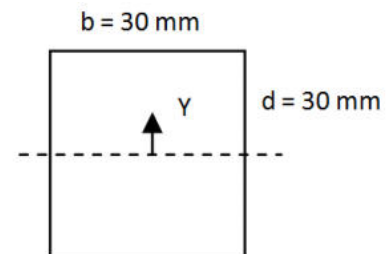
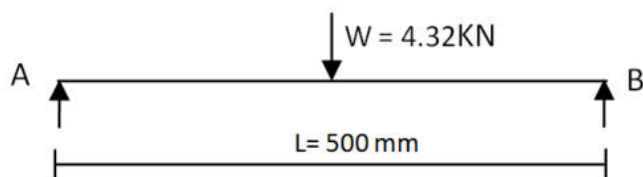
$$\therefore w = 300 \text{ N/mm}$$

$$\therefore w = 300 \text{ kN/m}$$

Example 2 A cast iron test beam 30 mm square in cross-section 500 mm long is simply supported at end. It fails at central point load at 4.32 KN. What load at free end will cause the failure of cantilever beam of 1 m span made of same material 30 mm x 60 mm in cross section?

Answer:

(1) For simply supported beam



$$I_{xx} = \frac{d^4}{12} = \frac{b^4}{12} = \frac{30^4}{12} = 67500 \text{ mm}^4$$

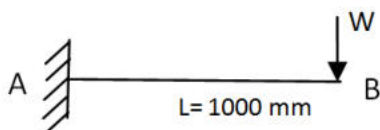
$$y = \frac{d}{2} = \frac{30}{2} = 15 \text{ mm}$$

$$M = \frac{wl}{4} = \frac{4.32 \times 10^3 \times 500}{4} = 540000 \text{ N} \cdot \text{mm}$$

$$\text{flexure eq}^n \quad \frac{M}{I} = \frac{\sigma}{Y}$$

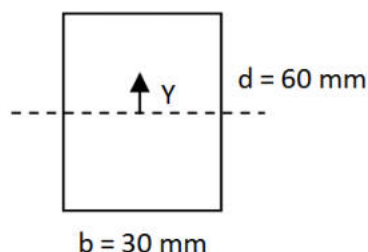
$$\therefore \frac{540000}{67500} = \frac{\sigma}{15} \quad \therefore \sigma = 120 \text{ N/mm}^2$$

(2) For cantilever beam



$$I_{xx} = \frac{bd^3}{12} = \frac{30 \times 60^3}{12} = 540 \times 10^3 \text{ mm}^4$$

$$y = \frac{d}{2} = \frac{30}{2} = 15 \text{ mm}$$



flexure eqⁿ $\frac{M}{I} = \frac{\sigma}{Y} \quad \therefore \frac{M}{540 \times 10^3} = \frac{120}{30}$
 $\therefore M = 2.16 \times 10^6 \text{ N} \cdot \text{mm}$

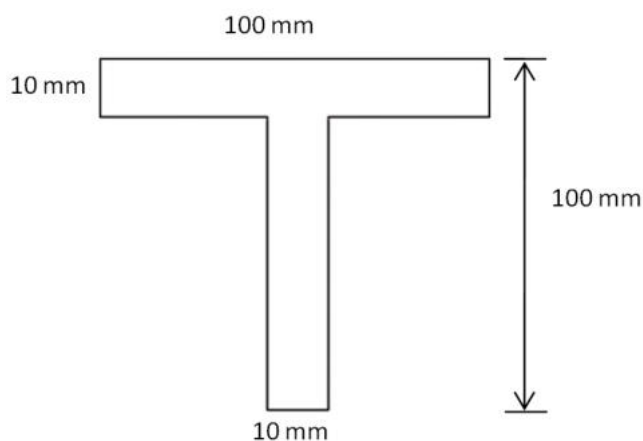
→ For cantilever $M = w \cdot l$

$$\therefore w = \frac{2.16 \times 10^6}{1000} = 2160 \text{ N}$$

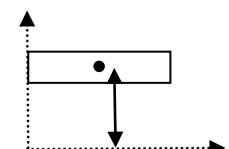
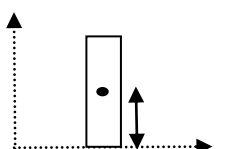
$$\therefore w = 2.16 \text{ KN}$$

Example 3: A section of beam as shown in fig. is subjected to a bending stress of 10 kN.m about the major axis. Draw bending stress distribution across the section.

Answer:



(1) Center of a given beam

Sr.no	Shape	Area (mm ²)	Y (mm)	AxY (mm ³)
1		A ₁ = 100 × 10 A ₁ = 1000	Y ₁ = 90 + $\frac{10}{2}$ Y ₁ = 95	A ₁ Y ₁ = 95000
2		A ₂ = 90 × 10 A ₂ = 900	Y ₂ = $\frac{90}{2}$ Y ₂ = 45	A ₂ Y ₂ = 40500
$\Sigma A = 1900$		$\Sigma AY = 135500$		

$$\bar{Y} = \frac{\Sigma AY}{\Sigma A} = \frac{135500}{1900}$$

$$\bar{Y} = 71.32 \text{ mm}$$

$$Y_b = 71.32 \text{ mm}$$

$$Y_t = 100 - 71.32$$

$$Y_t = 28.68 \text{ mm}$$

(2) Moment of inertia about centroidal horizontal axis

Sr.no.	Area	h (mm)	Ah ² (mm ⁴)	I _G or I _{self} (mm ⁴)	I _{xx} = I _G + Ah ² (mm ⁴)
1	A ₁ = 1000	h ₁ = Y _t - $\frac{d_1}{2}$ = 23.68	A ₁ h ₁ ² = 5607242.34	I _{G1} = $\frac{100 \times 10^3}{12}$ = 8333.34	I ₁ = 569075.7
2	A ₂ = 900	h ₂ = Y _b - $\frac{d_2}{2}$ = 26.32	A ₂ h ₂ ² = 623468.16	I _{G2} = $\frac{10 \times 90^3}{12}$ = 607500	I ₂ = 1230968.16

(3) Bending stress

$$M = 10 \text{ kN} = 10 \times 10^3 \times 10^3 \text{ N} \cdot \text{mm} = 10 \times 10^6 \text{ N} \cdot \text{mm}$$

$$I_{xx} = 1.8 \times 10^6 \text{ mm}^4$$

$$Y_{\text{top}} = 100 - 71.32 = 28.68 \text{ mm}$$

$$Y_{\text{bottom}} = 71.32 \text{ mm}$$

Bending stress at top

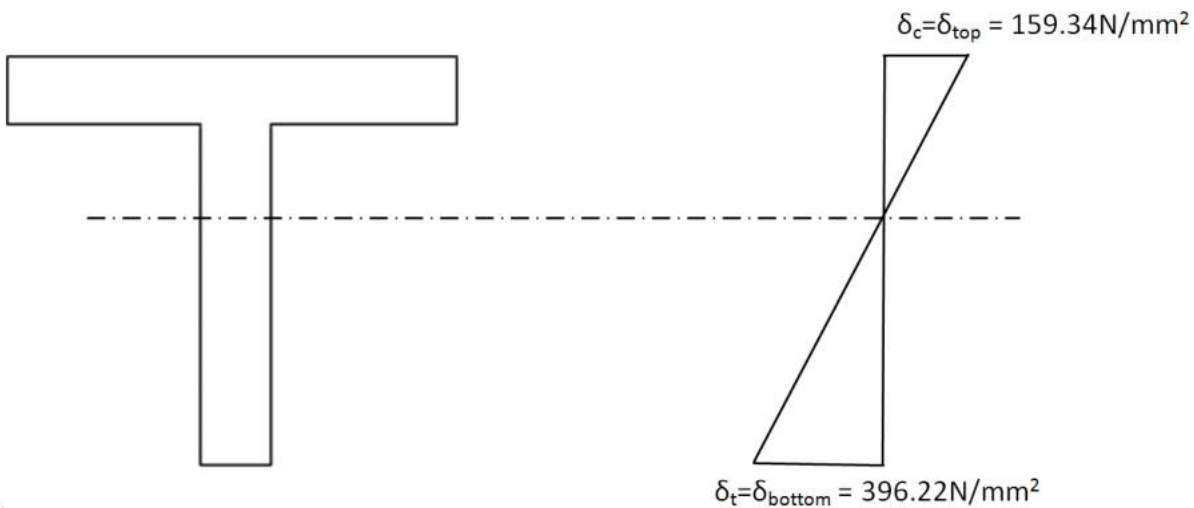
$$\frac{M}{I} = \frac{\sigma}{Y_t} \quad \therefore \frac{10 \times 10^6}{1.8 \times 10^6} = \frac{\sigma}{28.68}$$

$$\therefore \sigma_{\text{top}} = 159.34 \text{ N/mm}^2$$

Bending stress at bottom

$$\frac{M}{I} = \frac{\sigma}{Y_b} \quad \therefore \frac{10 \times 10^6}{1.8 \times 10^6} = \frac{\sigma}{71.32}$$

$$\therefore \sigma_{\text{bottom}} = 392.22 \text{ N/mm}^2$$



➤ **Shear stress distribution across beam section** (equation of shear stress)

Consider a small portion ABCD of length δx of a beam loaded with U.D.L as shown in fig (a).

Shear force (S.F) & Bending moment (B.M) will change at every point along the length of the beam.

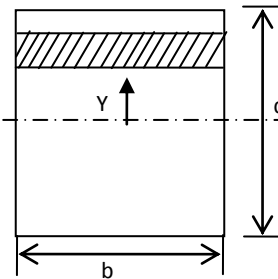
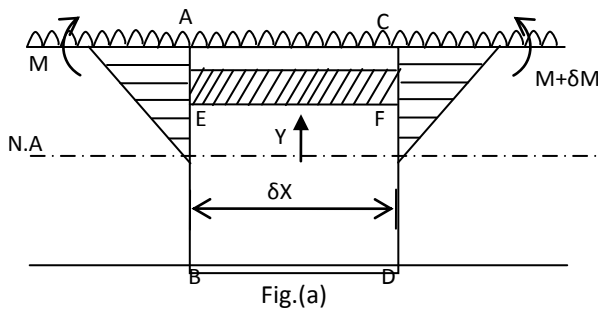
Let, $M = \text{B.M at AB}$

$M + \delta M = \text{B.M at CD}$

$F = \text{S.F at AB}$

$F + \delta F = \text{S.F at CD}$

$I = \text{M.I. of the section about the neutral axis.}$



Now consider an element strip EF at distance Y from the neutral axis.

$a = \text{cross section area of strip EF.}$

$\Sigma = \text{bending stress at AB.}$

Now, using flexure eqⁿ $\frac{M}{I} = \frac{\sigma}{Y}$

$$\therefore \sigma = \frac{M}{I} \times Y$$

$$\text{similarly, } \sigma + \delta\sigma = \frac{(M + \delta M)}{I} \times Y \quad (\sigma + \delta\sigma = \text{bending stress across CD section})$$

$$\therefore \sigma = \frac{M}{I} \times Y$$

Force acting across CD

$$= (\sigma + \delta\sigma) \times a$$

$$= \left(\frac{M + \delta M}{I} \right) \times Y \times a \quad \text{--- (i)}$$

Force acting across AB

$$= \text{stress} \times \text{area}$$

$$= \sigma \times a$$

$$= \frac{M}{I} \times Y \times a \quad \text{--- (ii)}$$

∴ Net unbalanced force on the strip

$$= \left\{ \frac{(M + \delta M)}{I} \times Y \times a \right\} - \left\{ \frac{M}{I} \times Y \times a \right\}$$

$$= \frac{\delta M}{I} \times Y \times a$$

Now, total unbalance force above neutral axis may be found out by integration above equation

between 0 & $\frac{d}{2}$

$$\therefore F = \int_0^{\frac{d}{2}} \frac{\delta M}{I} \times Y \times a \times dY$$

$$= \int_0^{\frac{d}{2}} \frac{\delta M}{I} (a \times Y) \times dY = \frac{\delta M}{I} \int_0^{\frac{d}{2}} a \times Y \times dY$$

$$F = \frac{\delta M}{I} \times a \bar{Y}$$

Where, A = area of beam above N.A

$\bar{Y}$ = distance between the centroid of the area & N.A

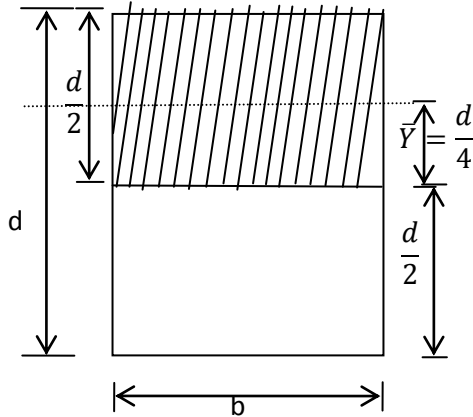
Now, shear stress = $\frac{\text{total shear force}}{\text{Area}}$

$$= \frac{\frac{\delta M}{I} \times a \bar{Y}}{\delta x \cdot b} \quad \text{substituting}$$

$$= \frac{\delta M}{\delta x} \times \frac{a \bar{Y}}{I \cdot b} \quad \frac{\delta M}{\delta x} = F = S.F$$

$$\tau = \frac{F \cdot A \cdot \bar{Y}}{I \cdot b}$$

- **Prove that the maximum shear stress in a rectangular section of beam is 1.5 times of average shear stress.**



Consider rectangular beam as shown in fig.

b = width, d = depth

(i) Average shear stress:-

We know that average shear stress is given as

$$\tau_{avg} = \frac{F}{b \times d}$$

(ii) Maximum shear stress:-

Maximum shear stress occurs at neutral axis which lies at $\frac{d}{2}$ from top & bottom.

$$\tau_{max} = \frac{F A \bar{Y}}{I b}$$

$$\text{here, } A = d \times \frac{d}{2} = \frac{bd}{2}$$

$$\bar{Y} = \frac{d}{2}, \quad I = \frac{bd^3}{12}$$

$$\therefore \tau_{max} = \frac{F \times \frac{b \times d}{2} \times \frac{d}{4}}{\frac{bd^3}{12} \times b} = \frac{3}{2} \frac{F}{bd}$$

$$\therefore \frac{\tau_{max}}{\tau_{avg}} = \frac{\frac{3}{2} \times \frac{F}{b \times d}}{\frac{F}{b \times d}} = \frac{3}{2} = 1.5$$

$$\therefore \tau_{max} = 1.5 \tau_{avg}$$

Prove that the maximum shear stress in a circular section of beam is 4/3 times of average shear stress.

→ Consider a plane circular lamina having diameter 'd'

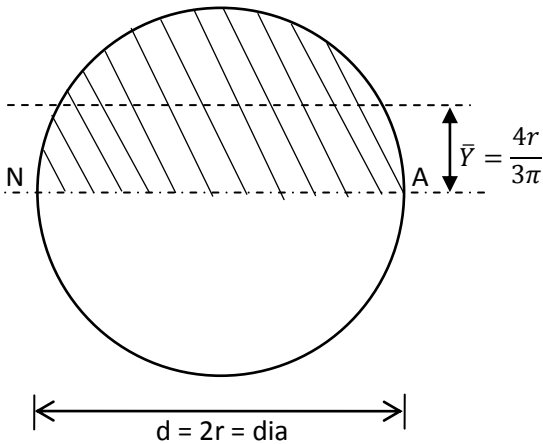
(i) Average shear stress:-

$$\tau_{avg} = \frac{F}{A} = \frac{F}{\frac{\pi d^2}{4}} = \frac{4F}{\pi d^2}$$

(ii) Maximum shear stress:-

Maximum shear stress lies at neutral axis which passes through its centroid.

The centroid of circle lies at $\frac{d}{2}$ from top & bottom.



$$\tau_{max} = \frac{F A \bar{Y}}{I b}$$

$$\text{here, } A = \frac{1}{2} \left(\frac{\pi d^2}{4} \right) = \frac{\pi d^2}{8}, \quad \bar{Y} = \frac{4r}{3\pi} = \frac{2d}{3\pi}$$

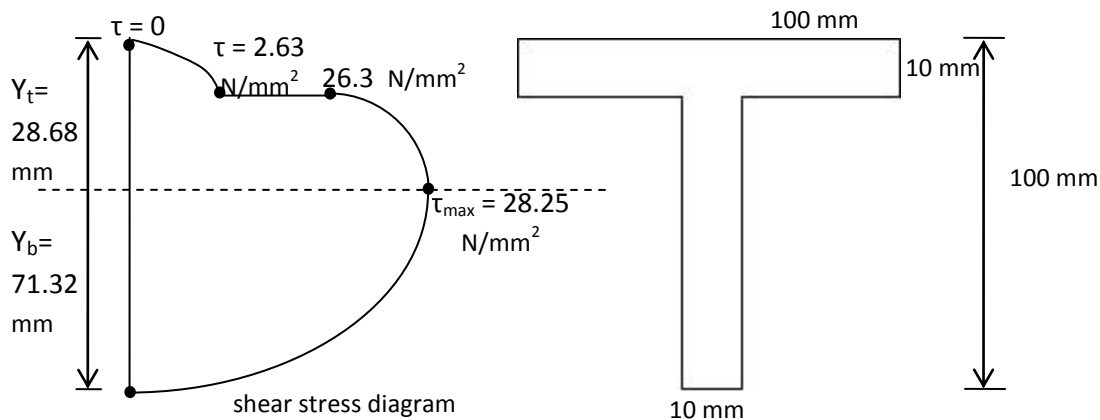
$$I = \frac{b d^3}{12}, \quad b = d$$

$$\therefore \tau_{max} = \frac{F \times \frac{\pi d^2}{8} \times \frac{2d}{3\pi}}{\frac{\pi d^4}{64} \times d} = \frac{16F}{3\pi \cdot d^2}$$

$$\text{Now, } \frac{\tau_{max}}{\tau_{avg}} = \frac{\frac{16F}{3\pi \cdot d^2}}{\frac{4F}{\pi \cdot d^2}} = \frac{4}{3} = 1.33$$

$$\therefore \tau_{max} = \frac{4}{3} \tau_{avg}$$

Example . A section of beam as shown in fig. is subjected to a shear force of 20KN. Draw shear stress distribution across the section.

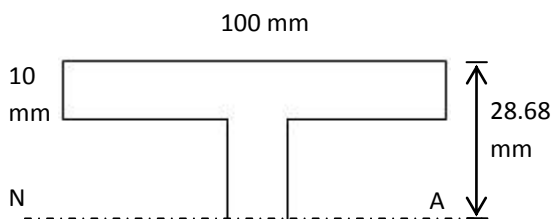


Ans:-

Shear force on beam = 20 KN

(i) shear stress at junction of flange & web.

$Y_{\text{top}} = 28.68 \text{ mm}$, $Y_{\text{bottom}} = 71.32 \text{ mm}$



$$A \bar{Y} = \{(100)(10)\} \times \left(28.68 - \frac{10}{2}\right) = 23680 \text{ mm}^3$$

$$\tau = \frac{F A \bar{Y}}{I \cdot b}, \quad I = 1.8$$

$$b = 100 \text{ mm}$$

$$\tau = \frac{F A \bar{Y}}{I \cdot b} = \frac{20 \times 10^3 \times 23680}{1.8 \times 10^6 \times 100} = 2.63 \text{ N/mm}^2$$

At the junction, there is sudden increase in the shear stress.

$$= 2.63 \times \frac{100}{10} = 26.3 \text{ N/mm}^2$$

(ii) Shear stress at C.G.

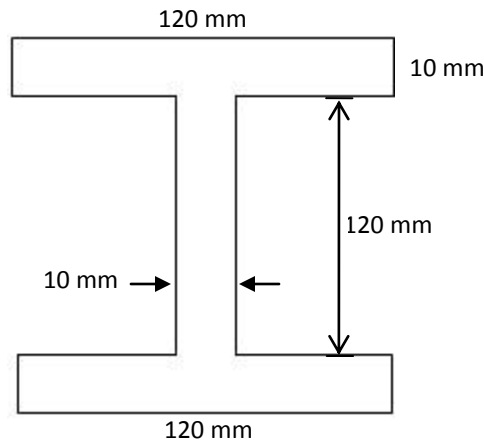
$$A \bar{Y} = \{(100 \times 10) \times (28.5 - 5)\} + \left\{ (18.68 \times 10) \times \frac{18.68}{2} \right\}$$

$$= 23680 + 1744.71$$

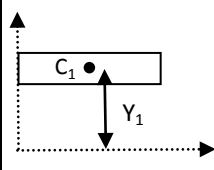
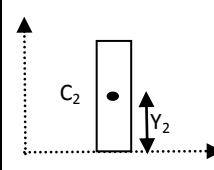
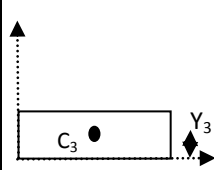
$$= 25424.712 \text{ mm}^3$$

$$\tau_{\text{max}} = \frac{F A \bar{Y}}{I \cdot b} = \frac{20 \times 10^3 \times 25424.712}{1.8 \times 10^6 \times 100} = 28.25 \text{ N/mm}^2$$

Ex- Fig. shows a beam cross section subjected to a shear force of 200 KN. Determine the shearing stress at neutral axis & sketch the shear stress distribution diagram.



Ans. (1) Center of a given section

Sr.no	Shape	Area (mm ²)	Y (mm)	A*Y (mm ³)
1		A ₁ =10*120 A ₁ =1200	Y ₁ = 10+120+ $\frac{10}{2}$ Y ₁ = 135	A ₁ Y ₁ = 162000
2		A ₂ =10*120 A ₂ =1200	Y ₂ =10+ $\frac{120}{2}$ Y ₂ =70	A ₂ Y ₂ = 84000
3		A ₃ =10*120 A ₃ =1200	Y ₃ = $\frac{10}{2}$ Y ₃ =5	A ₃ Y ₃ = 6000
$\sum A = 3600$		$\sum AY = 252000$		

$$\bar{Y} = \frac{\sum AY}{\sum A} = \frac{252000}{3600}$$

$$\therefore \bar{Y} = Y_b = Y_t = \frac{Y}{2}$$

$$= 70 \text{ mm}$$

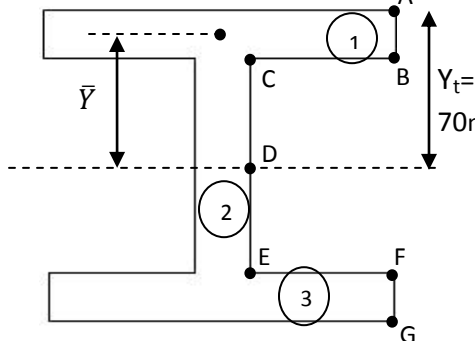
(2) Moment of inertia about centroidal horizontal axis.

Sr.no.	Area	h (mm)	Ah ² (mm ⁴)	I _G or I _{self} (mm ⁴)	I _{xx} = I _G + Ah ² (mm ⁴)
1	A ₁ =1200	h ₁ =Y _t - $\frac{d_1}{2}$ =65	A ₁ h ₁ ² =5.07X10 ⁶	I _{G1} = $\frac{b_1 \times d_1^3}{12}$ =1X10 ⁴	I ₁ = 5.08X10 ⁶
2	A ₂ =1200	h ₂ =Y _t - Y ₂ = 0	A ₂ h ₂ ² =0	I _{G2} = $\frac{b_2 \times d_2^3}{12}$ =1.44X10 ⁶	I ₂ = 1.44X10 ⁶
3	A ₃ =1200	H ₃ =Y _b - $\frac{d_3}{2}$ = 65	A ₃ h ₃ ² =5.07X10 ⁶	I _{G2} = $\frac{10 \times 90^3}{12}$ =1X10 ⁴	I ₂ = 5.08X10 ⁶

$$I_{xx} = I_1 + I_2 + I_3 = 11.8 \times 10^6 \text{ mm}^4$$

(3) Shear stress

Shear stress at the junction of flange & web



$$F = 14.5 \text{ KN}, \quad \tau = \frac{F A \bar{Y}}{I b}$$

$$A \bar{Y} = (120 \times 10) \times \left(Y_t - \frac{10}{2} \right) = 78000 \text{ mm}^3$$

$$I_{xx} = 11.6 \times 10^6 \text{ mm}^4, \quad b = 120 \text{ mm}$$

$$\tau_B = \frac{F A \bar{Y}}{I b_1} = \frac{14.5 \times 10^3 \times 78000}{11.6 \times 10^3 \times 120}$$

$$\tau_B = 0.8125 \text{ N/mm}^2$$

Now, Shear stress at C point.

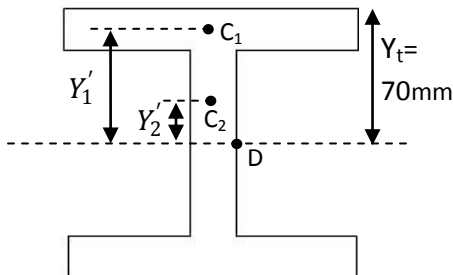
$$\therefore F = 14.5 \text{ KN}, \quad A \bar{Y} = 78000 \text{ mm}^3, \quad I_{xx} = 11.6 \times 10^6 \text{ mm}^4$$

$$b = b_2 = 10 \text{ mm} \quad (\because \text{in } \tau_B \text{ only width changes})$$

$$\tau_B = \frac{F A \bar{Y}}{I b_2} = \frac{14.5 \times 10^3 \times 78000}{11.6 \times 10^3 \times 10} \quad \text{Alternatively}$$

$$\tau_B = 9.75 \text{ N/mm}^2$$

Shear stress at neutral axis.



$$\tau_C = \tau_B \times \frac{b_1}{b_2} = \tau_B \times \frac{120}{10} = 9.75 \text{ N/mm}^2$$

$$\therefore F = 14.5 \text{ KN}, \quad I_{xx} = 11.6 \times 10^6 \text{ mm}^4, \quad b = b_2 = 10 \text{ mm}$$

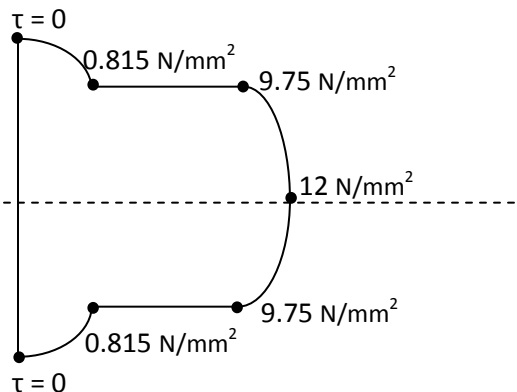
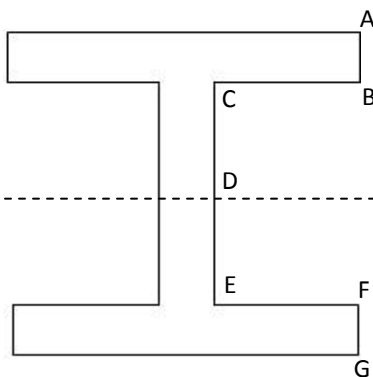
$$A \bar{Y} = A_1 Y_1' + \frac{A_2}{2} Y_2' = (120 \times 10) \left(70 - \frac{10}{2} \right) + \left(\frac{120 \times 10}{2} \right) \times \left(\frac{60}{2} \right)$$

$$A \bar{Y} = 96000 \text{ mm}^3$$

$$\tau_D = \frac{14.5 \times 10^3 \times 96000}{11.6 \times 10^3 \times 10} = 12 \text{ N/mm}^2$$

From geometry of a given section is symmetric about centroidal axis (Horizontal & Vertical)

$$\therefore \tau_A = \tau_G, \quad \tau_B = \tau_F, \quad \tau_C = \tau_E$$



shear stress distribution diagram

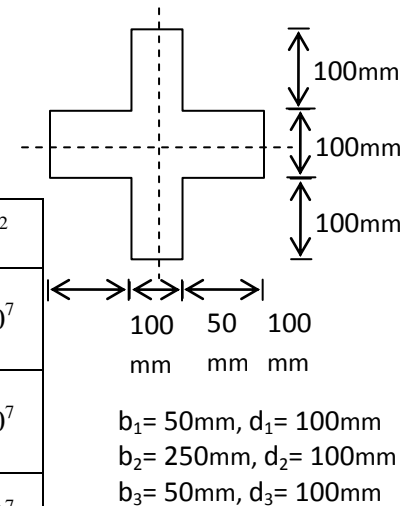
Example As shows figure a cross section subjected to a shearing force of 200 kN. Determine the shearing stress at neutral axis & sketch the shear stress distribution diagram across the section.

Ans. Given section is symmetric about Horizontal & Vertical axis

$$\therefore Y_{\text{top}} = Y_{\text{bottom}} = \bar{Y} = 150 \text{ mm}$$

Moment of inertia about centroidal horizontal axis.

S _n	Area	h (mm)	Ah ² (mm ⁴)	I _G or I _{self} (mm ⁴)	I _{xx} = I _G + Ah ²
1	A ₁ = 50 × 100 = 5000	h ₁ = Y _t - $\frac{100}{2}$ = 100	A ₁ h ₁ ² = 5 × 10 ⁷	I _{G1} = $\frac{b_1 \times d_1^3}{12}$ = 4.167 × 10 ⁶	I ₁ = 5.4167 × 10 ⁷
2	A ₂ = 100 × 250 = 25000	h ₂ = 0	A ₂ h ₂ ² = 0	I _{G2} = $\frac{b_2 \times d_2^3}{12}$ = 2.0834 × 10 ⁷	I ₂ = 2.0834 × 10 ⁷
3	A ₃ = 5000	H ₃ = Y _b - $\frac{100}{2}$ = 100	A ₃ h ₃ ² = 5 × 10 ⁷	I _{G2} = 4.167 × 10 ⁶	I ₂ = 5.4167 × 10 ⁷



Now, I_{xx} = I₁ + I₂ + I₃ = 129.167 × 10⁶ mm⁴

→ Stress at point B (shear)

A = 50 × 100 = 5000 mm², F = 200 kN

$$\bar{Y} = Y_t - \frac{d_1}{2} = 100 \text{ mm}, b = b_1 = 50 \text{ mm}$$

$$\tau_B = \frac{F A \bar{Y}}{I b} = \frac{200 \times 10^3 \times 5000 \times 100}{129.167 \times 10^6 \times 50}$$

$$\tau_B = 15.48 \text{ N/mm}^2$$

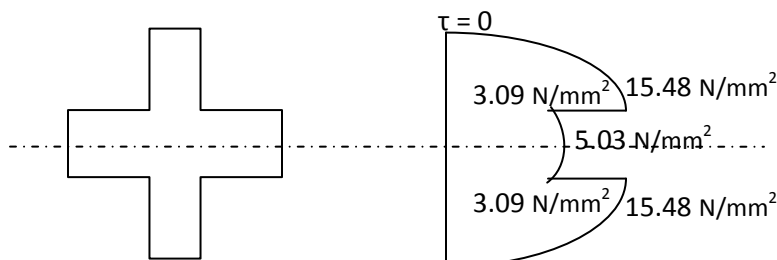
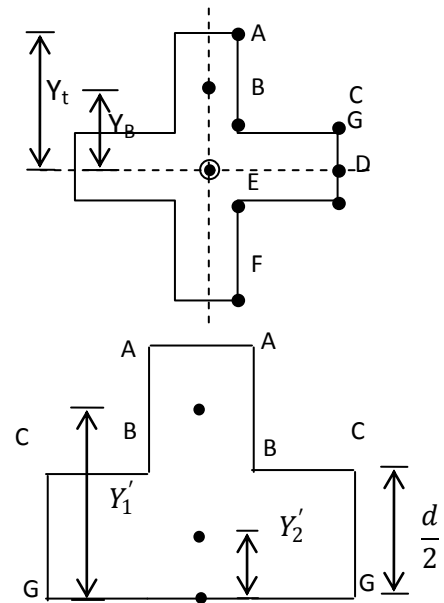
→ Stress at point C $\tau_C = \tau_B \times \frac{50}{250} = 3.09 \text{ N/mm}^2$

$$A = (b_1 \times d_1) \times \left(\frac{d_1}{2} + \frac{d_2}{2} \right) + \left(b_2 \times \frac{d_2}{2} \right) \times \left(\frac{d_2}{4} \right)$$

$$= A_1 Y'_1 + \frac{A_2}{2} \times Y'_2 = 812500 \text{ mm}^2$$

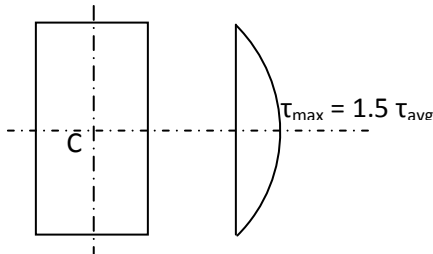
b = b₁ = 50mm

$$\tau_G = \frac{F A \bar{Y}}{I b} = \frac{200 \times 10^3 \times 812500}{129.167 \times 10^6 \times 250} = 5.03 \text{ N/mm}^2$$

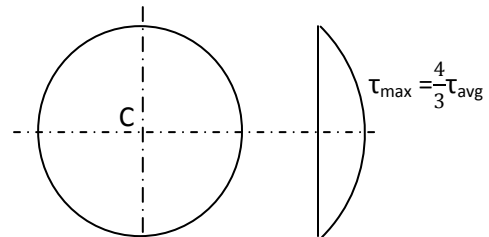


Draw shear stress distribution diagram for standard section

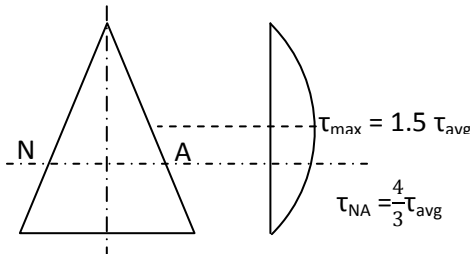
(1) Rectangular section



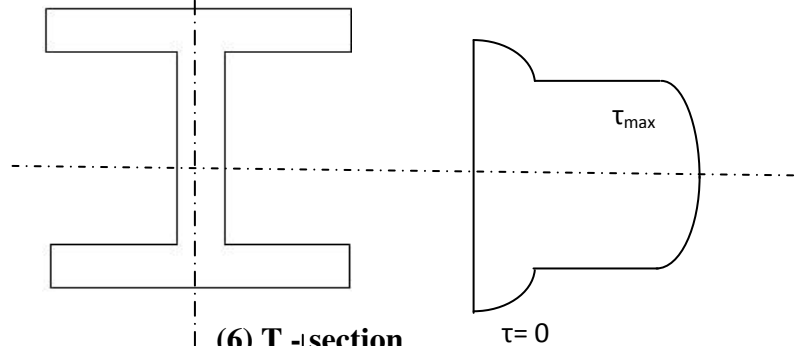
(2) Solid circular section



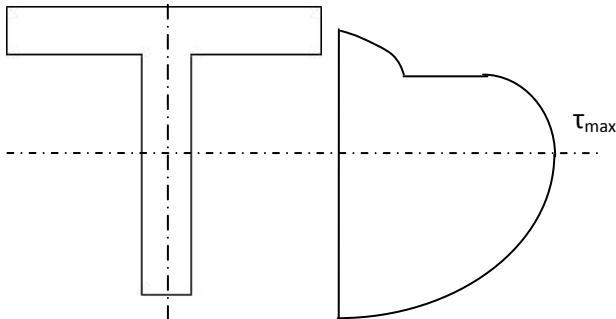
(3) Triangular section



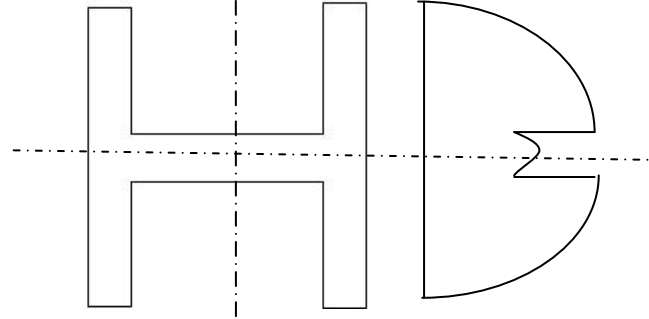
(4) I - section



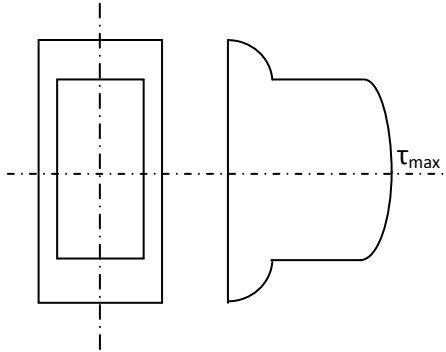
(5) T - section



(6) T - section



(7) Hollow rectangular section



(8) Solid square with diagonal horizontal (Rhombus)

